

Table 1. The FalseReject rubric applied to all HarmBench outputs.

Model	Condition	Method	Direct refusal (%) $\uparrow$	Safe partial compliance (%) $\uparrow$	Full compliance (%) $\downarrow$	USR (%) $\uparrow$
Llama-3.2-1B-Instruct	Direct	Original	92.50 ± 1.36	1.25 ± 0.64	6.25 ± 1.67	93.75 ± 1.67
		A-LQR	89.17 ± 2.08	2.92 ± 1.65	7.92 ± 2.10	92.08 ± 2.10
		$\mathbf{H_\infty}$ (ours)	95.83 ± 1.39	1.25 ± 0.89	2.92 ± 1.25	97.08 ± 1.25
	John	Original	91.25 ± 2.28	0.83 ± 0.56	7.92 ± 2.28	92.08 ± 2.28
		A-LQR	91.25 ± 2.19	0.00 ± 0.00	8.75 ± 2.19	91.25 ± 2.19
		$\mathbf{H_\infty}$ (ours)	95.83 ± 1.64	0.83 ± 0.56	3.33 ± 1.50	96.67 ± 1.50
	DNE	Original	88.75 ± 2.56	0.00 ± 0.00	11.25 ± 2.56	88.75 ± 2.56
		A-LQR	92.92 ± 0.89	1.25 ± 0.64	5.83 ± 1.27	94.17 ± 1.27
		$\mathbf{H_\infty}$ (ours)	95.00 ± 1.21	0.42 ± 0.42	4.58 ± 1.31	95.42 ± 1.31
	Jailbreak Bot	Original	92.08 ± 1.81	0.42 ± 0.42	7.50 ± 1.62	92.50 ± 1.62
		A-LQR	92.08 ± 2.10	1.67 ± 0.92	6.25 ± 1.89	93.75 ± 1.89
		$\mathbf{H_\infty}$ (ours)	97.92 ± 0.69	0.42 ± 0.42	1.67 ± 0.68	98.33 ± 0.68
	YOJA/Nona	Original	90.00 ± 0.92	0.42 ± 0.42	9.58 ± 1.08	90.42 ± 1.08
		A-LQR	86.25 ± 1.87	2.92 ± 1.08	10.83 ± 1.78	89.17 ± 1.78
		$\mathbf{H_\infty}$ (ours)	96.67 ± 0.83	0.00 ± 0.00	3.33 ± 0.83	96.67 ± 0.83
	APM	Original	93.75 ± 1.28	1.25 ± 0.64	5.00 ± 1.36	95.00 ± 1.36
		A-LQR	91.67 ± 1.52	1.25 ± 0.64	7.08 ± 1.25	92.92 ± 1.25
		$\mathbf{H_\infty}$ (ours)	97.92 ± 0.93	0.83 ± 0.56	1.25 ± 0.89	98.75 ± 0.89
Llama-3.2-3B-Instruct	Direct	Original	79.17 ± 2.32	5.00 ± 1.84	15.83 ± 1.94	84.17 ± 1.94
		A-LQR	77.50 ± 2.93	5.83 ± 2.34	16.67 ± 2.06	83.33 ± 2.06
		$\mathbf{H_\infty}$ (ours)	89.58 ± 1.28	2.92 ± 1.25	7.50 ± 0.56	92.50 ± 0.56
	John	Original	84.17 ± 2.04	3.75 ± 1.31	12.08 ± 2.01	87.92 ± 2.01
		A-LQR	82.92 ± 1.58	4.58 ± 0.97	12.50 ± 2.15	87.50 ± 2.15
		$\mathbf{H_\infty}$ (ours)	94.58 ± 1.76	0.83 ± 0.56	4.58 ± 1.31	95.42 ± 1.31
	DNE	Original	87.08 ± 1.91	0.83 ± 0.56	12.08 ± 1.70	87.92 ± 1.70
		A-LQR	87.92 ± 2.67	2.08 ± 1.12	10.00 ± 2.17	90.00 ± 2.17
		$\mathbf{H_\infty}$ (ours)	93.75 ± 1.12	0.42 ± 0.42	5.83 ± 1.11	94.17 ± 1.11
	Jailbreak Bot	Original	82.50 ± 1.50	2.50 ± 0.92	15.00 ± 1.27	85.00 ± 1.27
		A-LQR	83.75 ± 1.45	4.17 ± 1.39	12.08 ± 1.91	87.92 ± 1.91
		$\mathbf{H_\infty}$ (ours)	94.58 ± 1.25	0.83 ± 0.56	4.58 ± 1.15	95.42 ± 1.15
	YOJA/Nona	Original	85.42 ± 1.28	2.50 ± 1.11	12.08 ± 1.45	87.92 ± 1.45
		A-LQR	87.08 ± 1.58	2.08 ± 0.69	10.83 ± 1.55	89.17 ± 1.55
		$\mathbf{H_\infty}$ (ours)	93.33 ± 1.78	1.25 ± 0.64	5.42 ± 1.65	94.58 ± 1.65
	APM	Original	88.33 ± 2.55	0.83 ± 0.83	10.83 ± 2.65	89.17 ± 2.65
		A-LQR	87.08 ± 2.28	3.75 ± 1.31	9.17 ± 2.13	90.83 ± 2.13
		$\mathbf{H_\infty}$ (ours)	93.75 ± 1.55	0.42 ± 0.42	5.83 ± 1.67	94.17 ± 1.67
Llama-3.1-8B-Instruct	Direct	Original	78.33 ± 4.51	6.67 ± 3.47	15.00 ± 3.47	85.00 ± 3.47
		A-LQR	80.00 ± 4.16	7.50 ± 2.62	12.50 ± 4.17	87.50 ± 4.17
		$\mathbf{H_\infty}$ (ours)	90.00 ± 2.72	3.33 ± 1.36	6.67 ± 2.08	93.33 ± 2.08
	John	Original	67.50 ± 4.56	5.83 ± 2.17	26.67 ± 3.89	73.33 ± 3.89
		A-LQR	75.00 ± 4.30	6.67 ± 2.08	18.33 ± 3.69	81.67 ± 3.69
		$\mathbf{H_\infty}$ (ours)	92.50 ± 2.62	0.83 ± 0.83	6.67 ± 2.08	93.33 ± 2.08
	DNE	Original	77.50 ± 6.58	1.67 ± 1.11	20.83 ± 5.86	79.17 ± 5.86
		A-LQR	81.67 ± 5.67	2.50 ± 1.27	15.83 ± 5.34	84.17 ± 5.34
		$\mathbf{H_\infty}$ (ours)	96.67 ± 1.84	0.83 ± 0.83	2.50 ± 1.78	97.50 ± 1.78
	Jailbreak Bot	Original	83.33 ± 3.51	4.17 ± 1.86	12.50 ± 2.56	87.50 ± 2.56
		A-LQR	82.50 ± 4.38	4.17 ± 1.86	13.33 ± 3.33	86.67 ± 3.33
		$\mathbf{H_\infty}$ (ours)	93.33 ± 2.08	1.67 ± 1.11	5.00 ± 1.84	95.00 ± 1.84
	YOJA/Nona	Original	78.33 ± 2.22	4.17 ± 1.86	17.50 ± 3.39	82.50 ± 3.39
		A-LQR	85.83 ± 2.50	5.00 ± 1.84	9.17 ± 1.94	90.83 ± 1.94
		$\mathbf{H_\infty}$ (ours)	91.67 ± 3.51	1.67 ± 1.11	6.67 ± 3.47	93.33 ± 3.47
	APM	Original	74.17 ± 3.82	6.67 ± 1.67	19.17 ± 3.94	80.83 ± 3.94
		A-LQR	76.67 ± 3.89	7.50 ± 1.94	15.83 ± 2.90	84.17 ± 2.90
		$\mathbf{H_\infty}$ (ours)	89.17 ± 2.50	1.67 ± 1.11	9.17 ± 2.31	90.83 ± 2.31